

FacePass Android Version Release Note V3.8.2

Release note of v3.8.0&&v3.8.2:

1. Update detection, living body, and recognition models, and add attribute models;
2. Added the recognize interface of the bottomless library, optimizing the 1:1 process of personal identification;
3. Added feedFrameRGBIR interface, removed irFilter interface, and optimized the binocular process;
4. Added the initLocalGroup interface to support the identification of the same message among multiple base libraries;
5. There are age/gender/mask attributes, new hair/beard/hat/glasses/skin tone attributes are added in this version,
6. Added general attribute processing pipeline logic, which can control different thresholds and whether to send identification according to attributes such as mask/hat/glasses/skin color
7. Remove the original mask model and the mask double-threshold pipeline, the original mask double-threshold can be processed by the above general attribute pipeline instead
8. Added detectFaces, a pure picture detection interface;
9. Added remote detection mode;
10. Optimize the storage space occupied by the addFace interface to import the base library map
11. Optimize rotation parameter configuration;
12. Optimize interface image input format, unified support for Bitmap and FacePassImage;
13. Optimize the speed of multi-person recognition;
14. Optimize the error code and status returned by the recognize interface;
15. Optimize the error code and status returned by the livenessClassify interface;
16. Optimize the authorization process of mcvSafe;
17. The compare interface uses a scientific threshold to compare scores
18. Add a picture static 1:N comparison compare1xN interface;
19. Fix the problem that the roll value is abnormal when the face of the input related interface is not correct;
20. Add 3000 small base library support, you need to select the corresponding small base library authorization

Serial number	Model type	Model description	Model file name	Recommended threshold
1	Detection model		Inbound inspection use: detector.ar m.E.bin send identificatio	no

			n detection use: detector_rect.arm.E.bin pf.lmk.arm.D.bin		
2	Recognition model	Mainl ine scene mask scene	feat2.arm.H.v1.0_1core.bin	Base library size	Recommende d threshold
				10W	71
				5W	70
				2W	68
				1W	66
				5K	65
				5K (1:N mask recognition)	69
3	Living body mode I (choo se one from four) (Monocular recommended GPU)	Monocular CPU	liveness.CPU.rgb.int8.E.bin	65	
		Binocular CPU	liveness.CPU.rgbir.int8.E.bin	60	
		Monocular GPU	liveness.GPU.rgb.E.bin	60	

	(CPU recommended for binocular)	Binocular GPU	liveness.GPU.rgbir.E.bin	80
4	Face Angle-Blur Degree Model		attr.pose_blur.align.av200.190630.bin	no
5	RC attribute model		attr.RC.gray.12M.arm.200229.bin	no
6	Occlusion model		occlusion.all_attr_configurable.occ.190816.bin	no
7	Age-gender model		attr.age_gender.surveillance.nnjie.av200.0.1.0.190630.bin	no
8	Smile model		attr.smile.mgf29.0.1.1.181229.bin	no

Release note v3.7.0:

1. Update detection, living body, and recognition models;
2. Optimize the recognition performance under masks, and support setting thresholds separately for scenes with and without masks. For specific usage methods, see the developer guide
 - Configuration interface update, searchExtThreshold is newly added to FacePassConfig as the mask threshold, and mouthOccAttributeEnabled needs to be turned on;
 - The recognition result interface is updated, and the searchExtThreshold mask threshold and the mouthOccAttr mask judgment result are added in FacePassRecognitionDetail;
3. Update the extractFeature interface to return the result FacePassExtractFeatureResultFacePassRect
4. Added getFeatureByFaceTokenfaceTokenfeature
5. Added getFacelImagePathfaceToken
6. The FacePassHandler model initialization is separated from the control switch, that is, as long as the model is set, it is loaded, and then the corresponding model switch can be used to control whether to use the model function;
7. Fix the memory leak problem of addFaceDetect interface;
8. Optimize and expand the parameter control range of the addFaceConfig interface;
9. Optimize the hot switching function of live CPU and GPU models, and support various switching combinations of single and binocular live bodies;
10. Optimize the thumbnail logic of the inspection model;
11. Increase the recognition of topK interface, the parameter topK range limit, topK must be greater than 0;
12. Optimize the error code returned by recognition;
13. Optimization of other stability issues;

		Model description	Model file name	Recommended threshold	
1	Detection model		Inbound inspection use: detector.arm .D.bin send identification detection use: detector_rect.arm.D.bin pf.lmk.arm.D.bin	no	
2	Recognition model	Standard version (support Under the mask scene Recognition) (recommend)	feat2.arm.G.v1.0_1core.bin	Base library size	Recommended threshold
				10W	71
				5W	70
				2W	68
				1W	66
				5K	65
				1:N mask recognition supports 5K bottom library	Recommended threshold 64.71
				Recommended threshold for 1:1 person-to-document comparison 55.62	

3	Living model (Choose one from four)	Monocular CPU in vivo	liveness.CPU.rgb.int8.D.bin	60	
		Binocular CPU in vivo (Binocular recommendation)	liveness.CPU.rgbir.int8.D.bin	70	
		Live Monocular GPU (Monocular recommendation)	liveness.GPU.rgb.D.bin	60	
		Binocular GPU live body	liveness.GPU.rgbir.D.bin	80	
4	poseblur model		attr.pose_blur.align.av200.190630.bin		
5	Mouth occlusion attribute model		attribute.mouth.occ.gray.12M.190930.bin	no	
6	Occlusion model		occlusion.all_attr_configurable.oc.c.190816.bin	no	
7	Attribute AGE GENDER module type		attr.age_gender.surveillance.nnie.av200.0.1.0.190630.bin	no	
8	Smile model		attr.smile.mgf29.0.1.1.181229.bin	no	

Release note v3.6.0:

1. Update living body and recognition model;
2. Added a new mouth occlusion attribute model and corresponding pipeline to determine whether to wear a mask;
3. Package name change: Megvii.facepass becomes mcv.facepass, and the aar package FacePassAndroidSDK-MegSafe-release.aar is changed to FacePassAndroidSDK-McvSafe-release.aar;
4. Added setAffinity interface for binding core;

5. Newly added extract feature return landmark data interface extractFeatureWithLData;
6. Added a separate liveness detection interface livenessClassify;
7. Update the storage structure of face matting related interface in database;
8. Add the filtering of face occlusion (occlusion model needs to be enabled) and completeness of the related interface of storage;
9. Update the error code returned by the feedFrame and recognize interfaces;
10. Other function optimization;

		Model description	Model file name	Recommended threshold
1	Detection model		Inbound inspectio n use: detector. arm.C.bi n send identifica tion detection to use: detector_rect.arm.C.bin pf.lmk.float32.1015.bin	no

2	Recognition model	Standard version (supports recognition in mask scenarios) (recommend)	feat2.arm.F.v1.0.1core.bin	Base library size → mask threshold (thousands fp)1k 72.45 5k 72.45 10w 73.90 Base library size → common scene threshold (thousand five fp) 5k 71.97 1w 73.12 2w 74.10 5w 75.70 10w 76.59
		Dual-core version (does not support mask scene recognition)	feat2.arm.D.v1.0_2core.bin	Base library size → common scene threshold (thousand five fp) 5k 72.58 1w 73.80 2w 74.84 5w 76.49 10w 77.61
3	Living model (Choose one from four)	Monocular CPU in vivo	liveness.3288CPU.rgb.int8.C.bin	60
		Binocular CPU in vivo (Binocular recommendation)	liveness.3288CPU.rgbir.int8.C.bin	70
		Live Monocular GPU (Monocular recommendation)	liveness.3288GPU.rgb.C.bin	70
		Binocular GPU live body	liveness.3288GPU.rgbir.C.bin	80
4	poseblur model		attr.pose_blur.align.av200.190630.bin	
5	Mouth occlusion attribute model		attribute.mouth.occ.gray.12M.190930.bin	no
6	Occlusion model		occlusion.all_attr_configurable.occ.190816.bin	no
7	Attribute AGE GENDER module type		attr.age_gender-surveillance.nnjie.av200.0.1.0.190630. bin	no

8	Smile model		attr.smile.mgf29.0.1.1.181229.bin	no
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Release note v3.5.0:

1. Update detection, living body, and recognition models, and merge pose and blur models;
2. Update the FacePassConfig interface class, remove the redundant FacePassConfig construction interface, and update the input of the pose/blur model to poseBlurModel;
3. Live GPU model cache merge, supporting hot switching between binocular live and monocular live;
4. Added the clearAllGroupsAndFaces interface to clear all records in the base library;
5. Add recognize topK interface, return the recognition topK result;
6. Added the getLocalGroupFaceNum interface to return the number of faces in the base database;
7. Fix the small probability of crash caused by binocular live timeout and reset interface call;
8. Update the quality formula of the push picture;
9. Feature int8 quantification, improve search speed;
10. Update the living body preprocessing logic to improve the speed of living body detection;
11. Update the MegSafe authorization interface aar package and the corresponding demo;

		Model description	Model file name	Recommended threshold
1	Detection model		Inbound inspection use: detector.arm.C.bin send identification detection to use: detector_rect.arm.C.bin pf.lmk.float32.1015.bin	no
2	Recognition model	Standard version (recommended)	feat2.arm.C.v1.0_1core.bin	Base library size threshold 5k 74.03 1w 74.77 2w 76.47 5w 78.27 10w 79.73
3	Living model (Choose one from four)	Monocular CPU in vivo	liveness.3288CPU.rgb.int8.B.bin	40
		Binocular CPU in vivo (Binocular recommendation)	liveness.3288CPU.rgbir.int8.B.bin	60
		Live Monocular GPU (Monocular recommendation)	liveness.3288GPU.rgb.A.bin	50
		Binocular GPU live body	liveness.3288GPU.rgbir.B.bin	80

4	poseblur model		attr.pose_blur.align.av200.190630.bin	
5	Occlusion model		occlusion.all_attr_configurable.occ.190816.bin	no
6	Attribute AGE GENDER model		attr.age_gender-surveillance.nnue.av200.0.1.0.1.90630.bin	no
7	Smile model		attr.smile.mgf29.0.1.1.181229.bin	no

Release note v3.4.2:

1. Update living body and recognition model;
2. Update the MegSafe authorization interface aar package and the corresponding demo;
3. Please select the correct x86 model when using the x86 cloud feature to deliver.

The x86 matching model of the standard version of the recognition model fee.Aircraft.v3.0.cpu.1core.bin is named M_feature_x86_fp1025_v1.4.bin, and other recognition models do not support cloud feature delivery.

		Model description	Model file name	Recommended threshold
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1	Detection model		Use for storage detection: detector.client.Boita.bin Use for identification detection: detector_rect.client.Boita.bin lmk.face.low_resolution.gray.1M.pf.lmk8.190911.bin	no
2	Recognition model (Choose one of three)	Standard version (recommend)	feat.Aircraft.v3.0.cpu.1core.bin	75
		ID library version	feat.small.3288_rzj.115ms.int8.201901122.1core.bin	71.241
		Speed priority (Internal Beta Version)	feat2.Aircraft.fullface.v1.1.bin	74.8 (thousand five), 77.6 (one thousand)
3	Living model (Choose one from five)	Monocular CPU in vivo	liveness.3288CPU.rgb.int8.e2e.A.bin	60
		Binocular CPU in vivo (Binocular recommendation)	liveness.3288CPU.rgbir.int8.e2e.A.bin	80
		Live Monocular GPU (Monocular recommendation)	liveness.3288GPU.rgb.e2e.A.bin	40
		Binocular GPU live body	liveness.3288GPU.rgbir.e2e.A	80
		Binocular IR CPU Live (Internal Beta Version)	liveness.3288CPU.ir.20190910	80
4	pose model		pose.av200.20190630.bin	

5	blur model		blurness.av200.20190630.bin	
6	Occlusion model		occlusion.all_attr_configurable.occ.190816.bin	
7	Attribute AGE GENDER model		attr.age_gender.surveillance.nnue.av200.0.1.0.190630.bin	
8	Smile model		attr.blur.align.gray.general.mgf29.0.1.1.181229.bin	no

Release note v3.4.1:

Solve the problem that addFace and extractFeature interfaces detect abnormally large faces;

Release note v3.4.0:

model	Recommended threshold
feat.Aircraft.v3.0.cpu.1core.bin	75
feat.Aircraft.v3.0.cpu.2core.bin	
feat2.Aircraft.fullface.v1.1.bin	74.8 (five thousand), 77.6 (one thousand)
feat.small.3288_rzj.mbv3_165ms.float.201901015.1core.bin	61.4
liveness.3288CPU.rgb.20191025.bin	60
liveness.3288GPU.rgb.20191025.bin	
liveness.3288CPU.rgbir.20191025.bin	80
liveness.3288GPU.rgbir.20191025.bin	

1. Update detection, postfilter, pose, blur, age_gender, living body, recognition model;
2. The original online soft license and chip license are merged into one aar package;
3. Fix the issue of abnormal quality information such as pose/blur returned when adding a face fails;
4. Repair the recognition process, operate the bottom library, the problem of a small probability of crash;
5. Fix the problem that the faceMinThreshold threshold setting does not take effect;
6. Fix the problem that the brightness threshold setting of feedFrame interface does not take effect;
7. Optimize ifilter interface, optimize algorithm scheduling, and improve algorithm execution speed;

Release note of v3.3.0 official version:

model	Recommended threshold
feat.small.3288.mbv3_125ms.int8.20190825.1core.bin	76
liveness.3288CPU.rgb.20190825.bin	60
liveness.3288CPU.RGBIR.20190825.bin	60
liveness.3288GPU.rgb.20190825.bin	50
liveness.3288GPU.RGBIR.20190825.bin	70

1. Update the living body and recognition model to improve the performance of the model;
2. Increase the occlusion model to support partial occlusion filtering of faces;
3. Increase the feature extraction interface, increase the feature storage interface, and use it with the x86 quality feature comparison SDK.
5. Add MegSafe new soft authorization scheme;
6. Solve the problem that the binocular live model crashes when multiple faces are detected simultaneously;
7. Optimize the feedFrame and IRFilter interfaces to enhance the filtering of low-quality test results;
8. Optimize and update badcase picture save function.

Release note of v3.2.0 official version:

model	Recommended threshold
feat.small.3288_255MFlops_int8_150ms.1core.20190625.bin	79.8
liveness.3288CPU.rgb.20190630.bin	70
liveness.3288CPU.rgbir.20190722.bin	40
liveness.3288GPU.rgb.20190630.bin	70
liveness.3288GPU.rgbir.20190731.bin	90

1. Update detection, living body, and recognition models to improve model performance and speed;
2. Added half-face filtering function, effectively reducing half-face misrecognition;
3. Fix the abnormal score of compare interface;
4. Fix the stability problem caused by the abnormal parameter value of FacePassConfig;
5. Optimize and update log output and control methods, and add the function of saving badcase pictures.

Release note of v3.1.0 official version:

Client:

1. Modify the internal logic of IRfilter to reset the track state of the filtered message (you can output the message again without calling resetMessage);
2. Fix the bug that the repeated id of (bindGroup) at the time of warehousing caused warehousing failure;
3. When the base library synchronization fails, the base library version number is set to 0, and the full amount will be synchronized during the next synchronization;
4. Add infrared binocular live model support.

Release note of the official version of v3.0.0:

Client:

Note: This version of the authorization is not compatible with v2.1.1 and previous versions (but compatible with v3.0.0-test) because of major optimizations. If you are already in the production environment, it is recommended to consult technical support for effective upgrade. If you encounter difficulties in billing, you can seek commercial help.

1. Replaced all algorithm models, greatly improving the algorithm effect;
2. Changed the construction method of FacePassConfig, now it is more convenient to use;
3. Added GPU live algorithm, GPU can be used for acceleration on RK3288 model;
4. Added single-face mode, which can only recognize one person at a time by turning on single-face mode in the gate scene.

Release note of v3.0.0-test beta version:

Client:

Note: This version of the authorization is not forward compatible because of major optimizations. If you are already in the production environment, it is recommended to consult technical support for effective upgrade. If you encounter difficulties in billing, you can seek commercial help.

1. Added the functions of age, gender and smile index.

v2.1.1 official version Release note:

Client:

1. Change the detection model from iou_tracker model to face model.

Release note of v2.1.0 official version:

Client:

1. The detection model is updated to iou_tracker model;
2. Optimize the recognition model and increase the client base library to 5k;
3. Optimize the live model to improve the performance and speed of the live model.

server:

1. UpdateLive body model to optimize live body performance and speed

Release note of v2.0.2 official version:

Client:

1. Fix the bug that getLocalGroupInfo will crash when the number of base libraries exceeds 500;
2. Fix the underlying implementation of facepass, there will be a very small probability of crash in the recognize interface before (the 2.0.1 version has not been encountered at present, only in the special version for the central control);
3. Optimized the authorization logic. After the authorization file is damaged, the authorization file will be automatically deleted (the user can no longer authorize correctly unless the authorization file is manually deleted after the authorization file is damaged);
4. The bug of megface 2.9.1 needs to be replaced with megface 2.9.2;
5. Add a new interface setAddFaceConfig, which is used for user-defined quality parameters of storage;

server:

1. Added parameter configuration in the service/config interface, and users can customize parameter thresholds to adjust the quality of images in the library.
2. In the service/recognize interface, new identification and living parameter threshold settings are added, so that different clients can configure the threshold parameters according to the actual environment.
3. API document modification: the request parameter format in the image download interface was changed to url parameter.
4. API documentation update: image update for test package deployment and official package deployment.

Release note of v2.0.1 official version:

Client:

1. Optimize Docker to 18.03 to fix the problem of service suspended animation under high load;
2. Fix several bugs

server:

1. Fixed the bug that in the demo project, clicking the button to switch online/offline mode is invalid;
2. Fixed the bug that the getVersion interface returned the wrong version number;
3. Fixed the bug that the threshold value of detail in the FacePassRecognitionResult returned by the decodeResponse interface does not reflect the threshold value actually used;
4. The value of face min in the addFace warehousing interface is now consistent with the minimum face required by the warehousing face quality;
5. In the addFace warehousing interface, if a part of the detected face is out of the screen, it will be considered that the quality does not meet the requirements. It will be considered that the quality does not meet the requirements.